Combined effect of External Treatment of Herbal Medicine and Tuina in Congenital Muscular Torticollis: Systematic Review and Meta-analysis

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Supplement File 1. Search strategy for each database

Mesh terms and free words

(1) English databases

Concept		Keywords
Participants	Mesh terms	torticollis
	Free words	wry neck, twisted neck, torticol*, cervical dystonia, CMT, congenital muscular torticollis
Intervention	Mesh terms	herbal medic*
	Free words	herb*, decoction, remed*, Chinese medic*, Korean medic*, Kampo, formul*, herbal drug*, herbal medicine*, plant*, Chinese drug*, Chinese prescrip*, Chinese materica*, traditional medic*, traditional Chinese medic*, East Asian traditional medic*, external, application

We reviewed the titles, abstracts, and full texts of all papers that were searched with intervention keywords up to the external treatment of herbal medicine, due to there was no data when searching add tuina related keywords.

(2) Chinese databases

Concept		Keywords
Participants	Free words	wry neck, twisted neck, torticollis, cervical dystonia, 扭脖子, 斜颈, 颈肌张力障碍
Intervention	Free words	Chinese herbal drugs, traditional Chinese medicine, Chinese herbal medicine, traditional medicine, external, decoction, extract, oil, herbal, herb, 中药, 中医药, 汤, 丸, 膏, 散, 方, 颗粒, 胶囊, 自拟, 外敷, 贴敷, 外治
	Free words	tuina, chuna, manipulation, manual, massage, acupressure, manipulate, 推拿, 手法, 按摩, 推法

(3) Korean databases

Concept	Keywords

Participants	Free words	사경, 선천성 근 사경, 선천성 근성 사경, 선천성 사경, torticollis, CMT, congenital muscular torticollis
Intervention	Free words	한약, 외용, 연고, 첩부, 훈증, 온열, herb, herbal medicine, external, ointment, application, fumigation, moxibustion, patch
	Free words	추나, 도수, 마사지, 이완, tuina, chuna

(1) Pubmed

#1 "torticollis"[Mesh terms]

#2 "wry neck"[Title/Abstract] OR "twisted neck"[Title/Abstract] OR "torticol*"[Title/Abstract] OR "cervical dystonia"[Title/Abstract] OR "CMT"[Title/Abstract] OR "congenital muscular torticollis"[Title/Abstract]

#3 #1 OR #2

#4 "herbal medic*" [Mesh terms]

#5 "herb*"[Title/Abstract] OR "decoction"[Title/Abstract] OR "remed*"[Title/Abstract] OR "Chinese medic*"[Title/Abstract] OR "Korean medic*"[Title/Abstract] OR "Kampo"[Title/Abstract] OR "formul*"[Title/Abstract] OR "herbal drug*"[Title/Abstract] OR "herbal medicine*"[Title/Abstract] OR "plant*"[Title/Abstract] OR "Chinese drug*"[Title/Abstract] OR "Chinese prescrip*"[Title/Abstract] OR "Chinese materica*"[Title/Abstract] OR "traditional medic*"[Title/Abstract] OR "traditional Chinese medic*"[Title/Abstract] OR "East Asian traditional medic*"[Title/Abstract] OR "external"[Title/Abstract] OR "application"[Title/Abstract]

#6 #4 OR #5

#7 #3 AND #6

(2) Excerpta Medica dataBASE

#1 'torticollis'/exp

#2 'wry neck':ab,ti OR 'twisted neck':ab,ti OR 'torticol*':ab,ti OR 'cervical dystonia':ab,ti OR 'CMT':ab,ti OR 'congenital muscular torticollis':ab,ti

#3 #1 OR #2

#4 'herbal medic*'/exp

#5 'herb*':ab,ti OR 'decoction':ab,ti OR 'remed*':ab,ti OR 'chinese medic*':ab,ti OR 'korean medic*':ab,ti OR kampo:ab,ti OR formul*:ab,ti OR 'herbal drug*':ab,ti OR 'herbal medicine*':ab,ti OR 'plant*':ab,ti OR 'extract*':ab,ti OR 'chinese drug*':ab,ti OR 'chinese prescrip*':ab,ti OR 'chinese materica*':ab,ti OR 'traditional medic*':ab,ti OR 'traditional chinese medic*':ab,ti OR 'east asian traditional medic*':ab,ti OR 'external':ab,ti OR 'application':ab,ti

#6 #4 OR #5

#7 #3 AND #6

(3) Cochrane Central Register of Controlled Trials

#1 MeSH descriptor: [torticollis] explode all trees

#2 wry neck OR twisted neck OR torticol* OR cervical dystonia OR CMT OR congenital muscular torticollis (Word variations have been searched)

#3 #1 OR #2

#4 MeSH descriptor: [herbal medic*] explode all trees

#5 herb* OR decoction OR remedy OR Chinese medic* OR Korean medic* OR Kampo OR formula* OR herbal drug* OR herbal medic* OR plant* OR Chinese drug* OR Chinese prescript* OR Chinese material* OR traditional medic* OR traditional Chinese medic* OR East Asian traditional medic* OR external OR application (Word variations have been searched)

#6 #4 OR #5

#7 #3 AND #6

(4) China National Knowledge Infrastructure

SU=('wry neck' + 'twisted neck' + 'torticollis'+ 'cervical dystonia' + '扭脖子' + '斜颈' + '颈肌张力障碍') AND SU=('Chinese herbal drugs' + 'traditional Chinese medicine' + 'Chinese herbal medicine' + 'traditional medicine' + 'external' + 'decoction' + 'extract' + 'oil' + 'herbal' + 'herb' + '中药' + '中医药' + '汤' + '丸' + '膏' + '散' + '方' + '颗粒' + '胶囊' + '自拟' + '外敷' + '贴敷' + '外治') AND FT=('tuina' + 'chuna' + 'manipulation' + 'manual' + 'massage' + 'acupressure' + 'manipulate' + '推拿' + '手法' + '按摩' + '推法')

(5) Chinese Scientific Journal Database

M=("wry neck" OR "twisted neck" OR "torticollis" OR "cervical dystonia" OR "扭脖子" OR "斜颈" OR "颈肌张力障碍") AND M=("Chinese herbal drugs" OR "traditional Chinese medicine" OR "Chinese herbal medicine" OR "traditional medicine" OR "external" OR "decoction" OR "oil" OR "herbal" OR "herb" OR "中药" OR "中医药" OR "汤" OR "丸" OR "膏" OR "散" OR "方" OR "颗粒" OR "胶囊" OR "自拟") AND M=("extract" OR"外敷" OR "贴敷" OR "外治") AND M=("tuina" OR "chuna" OR "manipulation" OR "manual" OR "massage" OR "acupressure" OR "manipulate" OR "推拿" OR "手法" OR "按摩" OR "推法")

(6) Wan Fang Database

主题:(wry neck or twisted neck or torticollis or cervical dystonia or 扭脖子 or 斜颈 or 颈肌张力障碍) and 主题:(Chinese herbal drugs or traditional Chinese medicine or Chinese herbal medicine or traditional medicine or external or decoction or oil or herbal or herb or 中药 or 中医药 or 汤 or 丸 or 膏 or 散 or 方 or 颗粒 or 胶囊 or 自拟) and 主题:(extract or 外敷 or 贴敷 or 外治) and 主题:(tuina or chuna or manipulation or manual or massage or acupressure or manipulate or 推拿 or 手法 or 按摩 or 推法)

(7) Oriental Medicine Advanced Searching Integrated System

#1 (전체=사경 AND 전체=외용 한약)

#2 (전체=torticollis) AND (전체=herbal medicine OR 전체=external OR 전체=application)

#3 #1 AND #2

(8) Korean Studies Information Service System

#1 (전체=사경 AND 전체=외용 한약)

#2 (전체=wry neck OR 전체=twisted neck OR 전체= torticollis OR 전체 = cervical dystonia) AND (전체=TCM OR 전체=herbal medicine OR 전체=decoction OR 전체=CHM OR 전체=traditional Chinese medicine OR 전체=Chinese herbal medicine OR 전체=Korean medicine OR 전체=Korean herbal medicine OR 전체=alternative medicine OR 전체=external OR 전체= application)

3 #1 AND #2

(9) Science On

#1 (전체=사경 AND 전체=외용 한약)

#2 (전체=wry neck OR 전체=twisted neck OR 전체= torticollis OR 전체 = cervical dystonia) AND (전체=TCM OR 전체=herbal medicine OR 전체=decoction OR 전체=CHM OR 전체=traditional Chinese medicine OR 전체=Chinese herbal medicine OR 전체=Korean medicine OR 전체=Korean herbal medicine OR 전체=alternative medicine OR 전체=external OR 전체= application)

3 #1 AND #2

(10) Research Information Sharing Service

#1 (전체: 사경 <AND> 전체: 외용 한약)

#2 (전체: wry neck <OR> 전체: twisted neck <OR> 전체: torticollis <OR> 전체: cervical dystonia) <AND> (전체: TCM <OR> 전체: herbal medicine <OR> 전체: decoction <OR> 전체: CHM <OR> 전체: traditional Chinese medicine <OR> 전체: Chinese herbal medicine <OR> 전체: Korean medicine <OR> 전체: Korean herbal medicine <OR> 전체: alternative medicine <OR> 전체: external <OR> 전체: application)

3 #1 AND #2